Factors Associated with Heart Disease: A Statistical Analysis of Clinical Data

1. Project Overview

Research Objective

To investigate demographic and clinical characteristics associated with the presence of heart disease using clinical data.

Dataset

UCI Heart Disease Dataset

Sample Size

920 patient records.

2. Data Quality Assessment

An initial assessment was conducted to examine the completeness, distribution and consistency of the dataset before statistical analysis.

2.1 Dataset Characteristics

The dataset contains 920 patient records from four data sources: Cleveland, Hungary, Switzerland, and VA Long Beach.

2.2 Missing Data

The initial assessment identified missing observations in several continuous variables. Table 1 summarises the number of valid and missing observations for each variable.

Table 1:
Summary of Valid and Missing Observations for Selected Variables

Variable	Valid Observations	Missing Observations
Age	920	0
Resting Blood Pressure (Sys)	861	59
Serum Cholesterol	890	30
Maximum Heart Rate	865	55
Exercise Induced ST depression	858	62
Fasting Blood Sugar	830	90

2.3 Potential Data Anomalies

The initial data screening identified the following values requiring consideration during data preparation:

- Resting Blood Pressure (Sys): One observation had a value of zero.
- Serum Cholesterol: 172 observations had values of zero.
- Exercise-induced ST depression: 12 observations had negative values.

These values were retained without replacement or imputation. Their implications were considered when interpreting the descriptive statistics and selecting variables for the final logistic regression analysis.

3. Data Cleaning

The dataset was reviewed for missing observations and unusual values before analysis. The original five-category heart disease diagnosis variable was retained, and a separate binary variable was created to distinguish patients with no heart disease from those with heart disease.

Missing fasting blood sugar observations were treated as missing and excluded from analyses involving this variable. Consequently, the bivariate analysis involving fasting blood sugar and the multivariable logistic regression were conducted using the available complete observations.

Zero values in resting blood pressure and serum cholesterol, as well as negative values in exercise-induced ST depression, were not replaced or imputed. These variables were not included among the predictors in the final logistic regression analysis.

No imputation of missing values was performed.

4. Descriptive Analysis

4.1 Demographic Characteristics

The dataset comprised 920 patient records. There were 726 males (78.9%) and 194 females (21.1%). The records were obtained from four data sources: Cleveland (304; 33.0%), Hungary (293; 31.8%), Switzerland (123; 13.4%) and VA Long Beach (200; 21.7%).

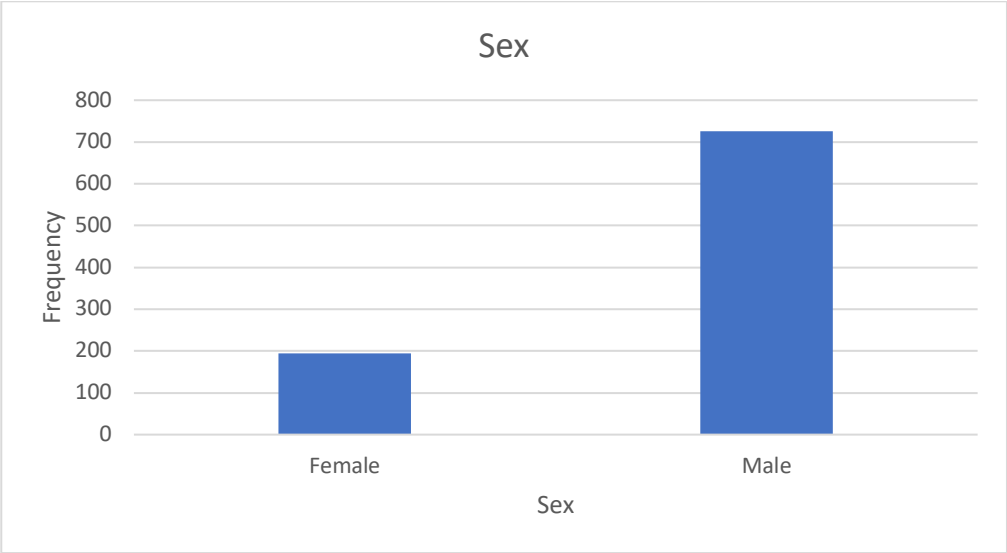


Figure 1:
Bar chart showing the distribution of patients by sex

The mean age of the patients was 53.51 years (SD = 9.43), with ages ranging from 28 to 77 years.

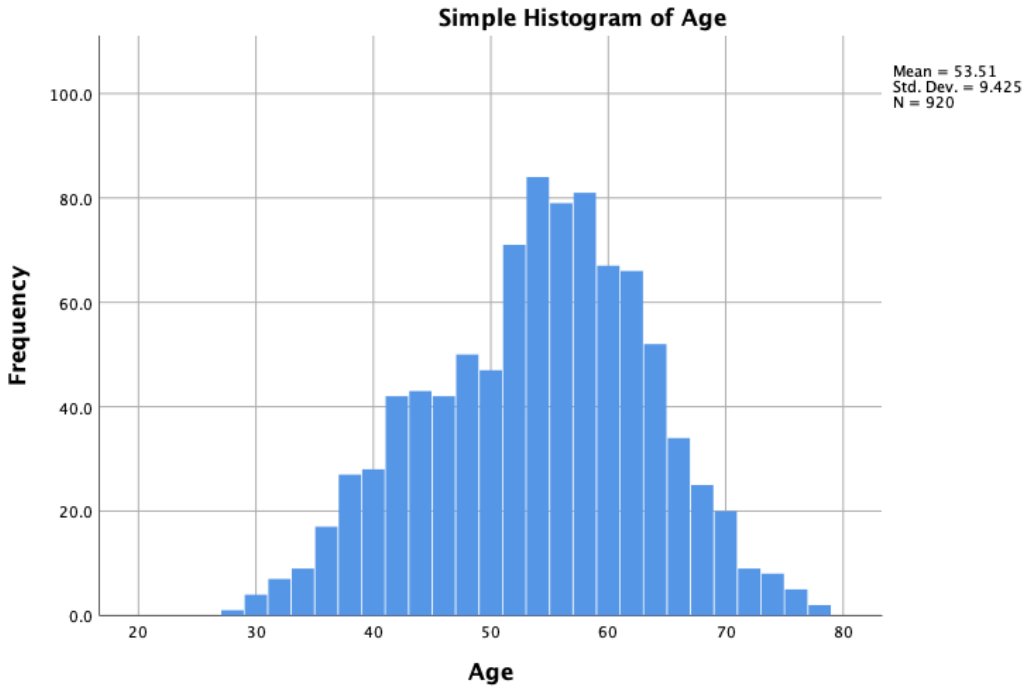


Figure 2:
Histogram showing the age distribution of patients

4.2 Clinical Characteristics

Regarding chest pain type, asymptomatic chest pain was the most frequently recorded category (496; 53.9%), followed by non-anginal pain (204; 22.2%), atypical angina (174; 18.9%) and typical angina (46; 5.0%).

Among observations with recorded fasting blood sugar results, 692 (83.4%) had values of 120 mg/dL or below, while 138 (16.6%) had values above 120 mg/dL.

For resting ECG results, 551 (60.0%) were normal, 188 (20.5%) showed left ventricular hypertrophy, and 179 (19.5%) showed ST-T abnormalities.

Exercise-induced angina was absent in 528 (61.0%) observations and present in 337 (39.0%) observations with recorded results.

For the slope of the peak exercise ST segment, flat was the most frequent category (345; 56.5%), followed by upsloping (203; 33.2%) and downsloping (63; 10.3%).

Among observations with recorded values for the number of major vessels visualized, 181 (58.6%) had zero vessels, 67 (21.7%) had one, 41 (13.3%) had two, and 20 (6.5%) had three.

For myocardial perfusion test results, 196 (45.2%) were normal, 192 (44.2%) showed reversible defects, and 46 (10.6%) showed fixed defects.

Heart disease diagnosis was classified into five categories. Of the 920 patients, 411 (44.7%) had no heart disease, 265 (28.8%) had category 1 heart disease, 109 (11.8%) had category 2 heart disease, 107 (11.6%) had category 3 heart disease, and 28 (3.0%) had category 4 heart disease.

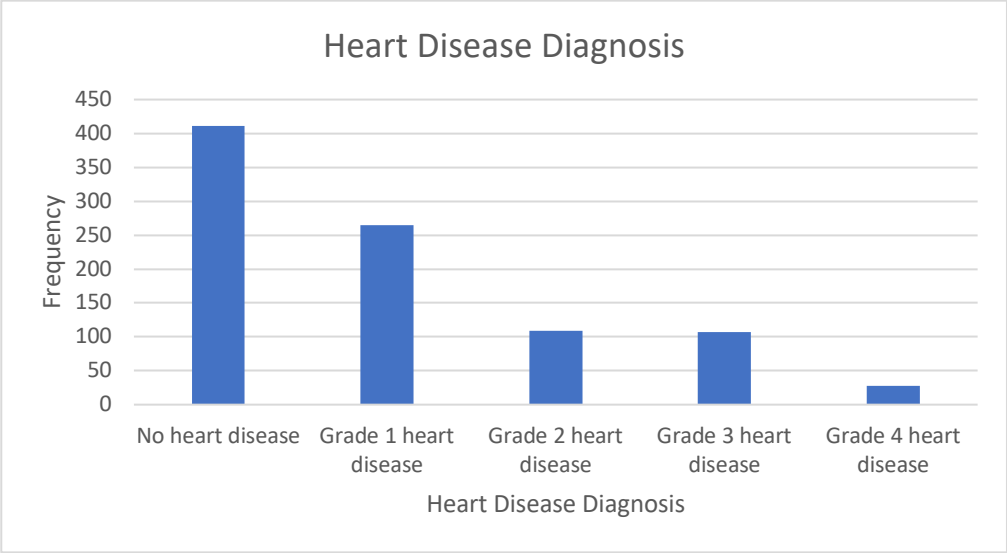


Figure 3:
Bar chart showing the distribution of heart disease diagnosis categories

4.3 Summary of Continuous Clinical Variables

Table 2 presents the descriptive statistics of the continuous demographic and clinical variables.

Table 2:
Descriptive Statistics of Continuous Demographic and Clinical Variables

Variable	Valid N	Mean	SD	Minimum	Maximum
Age (years)	920	53.51	9.43	28	77
Resting Blood Pressure (Sys)	861	132.13	19.07	0	200
Serum Cholesterol	890	199.13	110.78	0	603
Maximum Heart Rate	865	137.55	25.93	60	202
Exercise Induced ST depression	858	0.879	1.091	-2.6	6.2

4.4 Preparation of the Heart Disease Outcome Variable

The original Heart Disease Diagnosis variable contained five categories, coded from 0 to 4. For the subsequent analyses, it was recoded into a binary variable called Heart Disease Status. Patients with an original diagnosis value of 0 were classified as having no heart disease, while those with values from 1 to 4 were classified as having heart disease.

The resulting binary variable comprised 411 patients (44.7%) without heart disease and 509 patients (55.3%) with heart disease as shown in Table 3. The original five-category diagnosis variable was retained for reference.

Table 3:
Distribution of patients by heart disease status

Heart Disease Status	Frequency	Percentage
No heart disease	411	44.7%
Heart disease present	509	55.3%
Total	920	100.0%

5. Bivariate Analysis

5.1 Association Between Sex and Heart Disease Status

The chi-square test of independence revealed a statistically significant association between sex and heart disease status ($\chi^2 = 86.870$, df = 1, $p < 0.001$; N = 920).
Among female patients, 50 (25.8%) had heart disease, while 144 (74.2%) did not. Among male patients, 459 (63.2%) had heart disease, while 267 (36.8%) did not.

The proportion of patients with heart disease was higher among males than females. The distribution of heart disease status by sex is presented in Table 4.

Table 4:
Association Between Sex and Heart Disease Status

Sex	No heart disease, n (%)	Heart disease present, n (%)	Total
Female	144 (74.2%)	50 (25.8%)	194
Male	267 (36.8%)	459 (63.2%)	726
Total	411 (44.7%)	509 (55.3%)	920

5.2 Association Between Chest Pain Type and Heart Disease Status

The chi-square test of independence revealed a statistically significant association between chest pain type and heart disease status ($\chi^2 = 268.346$, df = 3, $p < 0.001$; N = 920). Among patients with asymptomatic chest pain, 392 (79.0%) had heart disease, while 104 (21.0%) did not. Among patients with atypical angina, 24 (13.8%) had heart disease, while 150 (86.2%) did not. Among patients with non-anginal pain, 73 (35.8%) had heart disease, while 131 (64.2%) did not. Among patients with typical angina, 20 (43.5%) had heart disease, while 26 (56.5%) did not.

The proportion of patients with heart disease varied across the chest pain categories, with the highest proportion observed among patients classified as asymptomatic. Table 5 presents the distribution of heart disease status across the chest pain categories.

Table 5:
Association Between Chest Pain Type and Heart Disease Status

Chest Pain Type	No Heart Disease, n (%)	Heart Disease Present, n (%)	Total
Asymptomatic	104 (21.0%)	392 (79.0%)	496
Atypical angina	150 (86.2%)	24 (13.8%)	174
Non-anginal	131 (64.2%)	73 (35.8%)	204
Typical angina	26 (56.5%)	20 (43.5%)	46
Total	**411 (44.7%)**	**509 (55.3%)**	**920**

5.3 Association Between Fasting Blood Sugar and Heart Disease Status

The chi-square test of independence revealed a statistically significant association between fasting blood sugar and heart disease status ($\chi^2 = 16.869$, df = 1, $p < 0.001$; N = 830). Among patients with fasting blood sugar ≤120 mg/dL, 339 (49.0%) had heart disease, while 353 (51.0%) did not. Among patients with fasting blood sugar >120 mg/dL, 94 (68.1%) had heart disease, while 44 (31.9%) did not.

The proportion of patients with heart disease was higher among those with fasting blood sugar >120 mg/dL. The association between fasting blood sugar and heart disease status is presented in Table 6.

Table 6:
Association Between Fasting Blood Sugar and Heart Disease Status

Fasting Blood Sugar	No Heart Disease, n (%)	Heart Disease Present, n (%)	Total
≤120 mg/dL	353 (51.0%)	339 (49.0%)	692
>120 mg/dL	44 (31.9%)	94 (68.1%)	138
Total	**397 (47.8%)**	**433 (52.2%)**	**830**

5.4 Comparison of Age by Heart Disease Status

An independent-samples t-test revealed a statistically significant difference in mean age between patients with and without heart disease ($t = -8.855$, $df = 845.862$, $p < 0.001$).

Patients with heart disease had a higher mean age (55.90 ± 8.72 years) than those without heart disease (50.55 ± 9.43 years), with a mean difference of 5.36 years (95% CI: 4.17–6.54). Table 7 presents the comparison of mean age between patients with and without heart disease.

Table 7.
Comparison of Mean Age by Heart Disease Status

Heart Disease Status	N	Mean Age (years)	Standard Deviation
No heart disease	411	50.55	9.43
Heart disease present	509	55.90	8.72

6. Multivariable Analysis

6.1 Factors Associated with Heart Disease

Binary logistic regression was used to examine the associations of age, sex, chest pain type, and fasting blood sugar with heart disease status, while accounting for the other characteristics. No Data Source variable was included.

The analysis included 830 patients with complete information for all four characteristics. Ninety patients (9.8%) were excluded because their fasting blood sugar results were missing.

The analysis showed that these characteristics, considered together, were significantly associated with heart disease status ($\chi^2 = 343.984$, $df = 6$, $p < 0.001$). The Nagelkerke R^2 was 0.453.

The Hosmer–Lemeshow test was not statistically significant ($\chi^2 = 7.431$, $df = 8$, $p = 0.491$), indicating no evidence of poor agreement between the regression estimates and the observed data based on this test.

6.2 Association of Individual Characteristics with Heart Disease

After accounting for the other characteristics, sex, chest pain type and age remained significantly associated with heart disease status. Fasting blood sugar was not significantly associated with heart disease status after this adjustment. The adjusted associations between the selected patient characteristics and heart disease status are presented in Table 8.

Table 8:
Adjusted Associations Between Patient Characteristics and Heart Disease

Patient characteristic	Adjusted odds ratio	95% confidence interval	p-value

Male vs Female	4.527	2.909–7.045	<0.001
Atypical angina vs Asymptomatic	0.063	0.038–0.106	<0.001
Non-anginal vs Asymptomatic	0.158	0.105–0.237	<0.001
Typical angina vs Asymptomatic	0.166	0.081–0.338	<0.001
FBS >120 vs ≤120 mg/dL	1.576	0.977–2.541	0.062
Age (per additional year)	1.060	1.039–1.081	<0.001

Reference categories: Female, asymptomatic chest pain and FBS ≤120 mg/dL. FBS = fasting blood sugar.

Male patients had approximately 4.5 times the adjusted odds of heart disease compared with female patients. Compared with patients classified as asymptomatic, those with atypical angina, non-anginal pain and typical angina had lower adjusted odds of heart disease.

Each additional year of age was associated with approximately 6% higher adjusted odds of heart disease.

Patients with fasting blood sugar above 120 mg/dL had higher estimated odds of heart disease than those with values of 120 mg/dL or below. However, this difference was not statistically significant after accounting for the other characteristics (p = 0.062).

6.3 Accuracy of Heart Disease Classification

Using a classification threshold of 0.50, the regression analysis correctly classified 77.3% of the patients. The classification results at a threshold of 0.50 are presented in Table 9.

Table 9:
Classification Accuracy of the Logistic Regression Analysis

Classification result	Correctly classified
Patients without heart disease	74.1%
Patients with heart disease	80.4%
All patients	77.3%

These results describe classification accuracy within the analysed dataset and do not establish how accurately the analysis would classify patients in a different population.

7. Regression Diagnostic Checks

7.1 Multicollinearity

Multicollinearity was assessed using tolerance and variance inflation factor (VIF) statistics. Tolerance values ranged from 0.937 to 0.978, while VIF values ranged from 1.022 to 1.067. These results indicated no concerning multicollinearity among the predictors.

7.2 Linearity of Age in the Logit

The linearity of age in the logit was assessed by including an interaction term between age and its natural logarithm in a diagnostic logistic regression analysis. The interaction term was not statistically significant ($p = 0.971$), providing no evidence of a departure from the linearity assumption for age.

8. Discussion

This analysis examined the relationship between selected demographic and clinical characteristics and the presence of heart disease among 920 patients in the UCI Heart Disease Dataset.

The findings showed that age and sex were significantly associated with heart disease status. Male patients had higher adjusted odds of heart disease than female patients. Increasing age was also associated with higher odds of heart disease, with each additional year associated with approximately 6% higher adjusted odds.

Chest pain type was significantly associated with heart disease status. In the adjusted analysis, patients in the atypical angina, non-anginal and typical angina categories had lower odds of heart disease than those classified as asymptomatic. This finding should be interpreted cautiously, as the dataset represents patients from selected clinical sources and the chest pain categories may reflect the characteristics of the patients referred for assessment.

Fasting blood sugar was significantly associated with heart disease status in the initial analysis. However, this association was no longer statistically significant after accounting for age, sex and chest pain type in the logistic regression analysis. This suggests that the initial association may be partly related to differences in these other patient characteristics.

Overall, the findings indicate that age, sex and chest pain type were independently associated with heart disease status in the analysed dataset.

9. Limitations

This analysis has several limitations.

1. Source of data: The dataset combines records from four clinical sources: Cleveland, Hungary, Switzerland and VA Long Beach. Differences in patient selection, clinical practice and data collection across these sources may have influenced the findings.
2. Missing data: Ninety patients (9.8%) were excluded from the logistic regression analysis because their fasting blood sugar results were missing. This reduced the number of patients included in the adjusted analysis.
3. Unusual recorded values: The initial data assessment identified zero values for resting blood pressure and serum cholesterol, as well as negative values for exercise-induced ST depression. These observations require cautious interpretation and may affect analyses involving these variables.
4. Observational data: The analysis examined associations and cannot establish that the identified characteristics caused heart disease.

5. Generalisability: The findings are based on a selected clinical dataset and should not be interpreted as estimates of heart disease prevalence or risk in the general population.

10. Conclusion

This analysis identified significant associations between heart disease status and age, sex and chest pain type. Older age and male sex were associated with higher adjusted odds of heart disease, while the chest pain categories showed different associations compared with the asymptomatic category. Fasting blood sugar was associated with heart disease in the initial analysis but was not statistically significant after adjustment for the other characteristics.

These findings provide insight into the relationships between selected patient characteristics and heart disease within the dataset. However, the limitations of the data should be considered when interpreting the results.